sonic boom
The thunderlike "bang" heard when an aircraft breaks through the speed of sound. This is caused by sudden changes in air pressure as the craft pushes air molecules out of its path.

spark plug
A device projecting into a combustion chamber that produces an electric spark to cause fuel ignition.

spoiler
An airplane control surface that, when activated, projects upward from the upper surface of a wing, increasing drag and decreasing lift. Spoilers are of different sizes, the largest being deployed just after touchdown to slow an aircraft and increase its grip on the runway.

squadron
A military unit of an air force. The term can refer both to flying and non-flying units. *See also* flight.

stabilizer
A surface that stabilizes an aircraft's flight. In most airplanes, the tail fin functions as the vertical stabilizer and the tailplane, as the horizontal stabilizer.

stall
When an aircraft stalls, it has the airflow over its wings disrupted so that turbulence is created and it loses lift. The aircraft's speed and the angle of attack of its wings are factors that affect whether stalling will occur. *See also* angle of attack.

static port
A small hole on the outside of an aircraft at a point where there is little disturbance to the airflow outside. It forms part of pressure-measuring systems used to determine an aircraft's airspeed.

stator
A nonmoving part of a turbine, usually in the form of a disk with shaped openings through which gases pass. It serves to direct gases onto the blades of a compressor turbine rotor efficiently. *See also* turbine.

Stealth Bomber
A bomber airplane that makes use of stealth technology. The name Stealth Bomber applies most commonly to the Northrop B-2 Spirit, a US aircraft in service since 1997. Stealth technology aims at making an aircraft invisible to radar and other modes of detection by such means as angular radar-deflecting surfaces and radar-absorbing materials. The technology has also been applied to some other military aircraft.

STOL
Abbreviation for short takeoff and landing, a desirable feature for fixed-wing aircraft that have to take off from locations such as clearings in the jungle. *See also* VSTOL.

strake
An additional fixed aerodynamic surface added to the fuselage of some aircraft to improve aerodynamic performance.

stratosphere
The layer of the earth's atmosphere above the troposphere, marked by a transition at which the air becomes warmer compared to that of the troposphere. The lack of turbulence and lower air density and resistance of the lower stratosphere have led to it becoming the preferred cruising location for airliners. *See also* troposphere.

stressed skin
A method of construction used in some aircraft fuselages in which a stretched flexible covering resists tensile (pulling) forces. Combined with a rigid framework, this creates a strong overall structure.

stroke
A single movement of a piston from the top to the bottom of a cylinder, and vice versa. *See also* piston engine.

Stuka
A German dive bomber of World War II. Short for "sturzkampfflugzeug," the German word for dive bomber. Although Stuka is not the name of an individual make of aircraft, it is commonly used to refer to the Junkers Ju 87, which was prominent in the early part of the war.

sump
An oil reservoir at the bottom of an engine.

supercharger
A type of air compressor in a piston engine that boosts power by increasing the amount of air fed to the cylinders. *See also* compressor.

supersonic
Faster than the speed of sound.

swept wing
A wing that is angled backward toward the rear of an airplane, in order to reduce drag. As it also reduces lift, it necessitates greater takeoff and landing speeds.

swing wing
The popular name for what is more technically called a variable geometry or variable sweep wing arrangement. The ability to adjust the horizontal angle of an aircraft's wings gives the combined advantages of lower drag at high speeds and higher lift at low speeds, but the mechanism itself adds to an aircraft's weight.

tail assembly
See empennage.

tail fin
The vertical part of an airplane's tail to which the rudder is attached.

taildragger
An airplane with a landing gear that consists of main wheels that are placed in front of the aircraft's center of gravity, together with a smaller wheel (or sometimes a skid) under the tail. Such aircraft sit on the ground with the nose pointing upward and the tail close to the ground—hence the name. When operating in rough terrain they offer the advantage that the propeller is further from the ground, hence reducing the risk of damage, during takeoff. At one time many large passenger aircraft were taildraggers, but now all of them have adopted the alternative tricycle gear configuration. *See also* tricycle gear.

tailplane
The horizontal part of an airplane's tail, in the shape of a small wing. It aids stability by controlling pitch, and includes the aircraft's elevators. *See also* pitch.

throttle
In aircraft, the cockpit lever that controls the amount of thrust generated by the engines. *See also* thrust.

thrust
The force that pushes a powered aircraft through the air.

torque
The twisting force created by a turning component, such as a propeller powered by an engine.

trailing edge
The rear edge of a wing, facing toward the back of the aircraft.

tricycle gear
An arrangement of wheeled landing gear that consists of a single nose wheel plus a pair of main wheels or wheel assemblies, placed behind the airplane's center of gravity. *See also* taildragger.

trim
In general, the balance that a particular aircraft adopts in steady flight. Trim can be adjusted using tabs built into the aircraft's flight control surfaces.

troposphere
The lowest and densest layer of the Earth's atmosphere, where air turbulence is at its greatest and most weather phenomena are generated. *See also* stratosphere.

turbine
In general, a rotating device that extracts energy from gases or liquids passing through it. *See also* gas turbine.

turbo
Abbreviation for turbocharger.

turbocharger
A device similar to a supercharger but powered by a turbine that uses the energy of an aircraft's exhaust gases. *See also* supercharger.

turbofan
An engine developed from the turbojet that is used in most modern airliners. In a turbofan, the turbine, as well as powering the compressor, also powers a large fan at the front, which directs some of the incoming air to bypass the combustion chambers and turbine. The cool unburned air mixes with the exhaust gases at the back of the engine, greatly reducing the noise produced.

turbojet
The original jet engine. Fuel burned in the stream of air passing through the engine creates a mix of hot gases under pressure. The gases expand and are accelerated out of the back of the engine, creating the thrust that pushes the aircraft forward. The engine requires a compressor to compress the incoming air before combustion, and so a turbojet is also a gas turbine, the turbine's power being used to turn the compressor blades.

turboprop
An engine similar in arrangement to a turbojet, except that the turbine extracts nearly all the energy from the hot gases and uses it to drive a propeller. Therefore, a turboprop does not actually use "jet power" to propel itself forward.

turtle deck
The curved skin surrounding an open cockpit or cockpits, generally made from plywood or aluminum sheet.

two-stroke engine
A type of piston engine in which the cycle of operation—introducing the fuel to the cylinder, burning it, and getting rid of exhaust gases—requires only two strokes of the piston. *See also* four-stroke engine.

type rating
A license or qualification required to fly an aircraft of a particular type, in addition to general flying qualifications. The need for such additional qualifications varies widely between different countries.

undercarriage
The landing gear of an aircraft—particularly wheeled landing gear for land aircraft. Wheeled undercarriages are often retractable into the fuselage or the wings to reduce drag in flight. *See also* landing gear.

unmanned aerial vehicle (UAV)
A powered, pilotless aircraft that can be controlled either remotely or, less commonly, by systems in the aircraft itself. The term usually excludes model airplanes used for recreation. A more informal name for a UAV is a "drone," and these are also called remotely piloted vehicles (RPVs). UAVs avoid risking pilots' lives and are used for military reconnaissance and targeted ground attack, as well as an increasing variety of civilian purposes.

VNE (velocity never exceeded)
The specific airspeed that, for safety reasons, a particular aircraft must not exceed.

VSI (vertical speed indicator)
An instrument that tells a pilot whether an aircraft is rising or falling in altitude.

VSTOL
Short for vertical and/or short takeoff and landing. Part of the significance of the phrase comes from the example of aircraft such as the Harrier "jump jet," which, while able to take off vertically using the rotatable engine nozzles, also has the facility to take off more economically, and also carry a heavier load, if a short runway is available.

VTOL
Short for vertical takeoff and landing. Although all helicopters have this ability, the term is usually used for fixed-wing aircraft such as the British Harrier "jump jet."

wing
The main aerodynamic lifting structure of an aircraft. There are many aircraft wing configurations, including low (projecting from near the bottom of the fuselage) and shoulder (projecting from near the top of the fuselage). Also, it is the name for an administrative subdivision of an air force. *See also* delta wings, parasol wing, swept wing, swing wing.

yaw
The tendency for an airplane to swing from side to side horizontally during flight. It is controlled by adjusting the aircraft's rudder and other control surfaces.